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PREPREG COMPOSITE MATERIAL PLY AND BACKING SEPARATION SYSTEMS AND METHODS

Abstract

A system for separating a prepreg composite material ply from a backing includes a first support configured to contact a first surface of the backing apart from a corner defined by the ply and the backing. A second support is configured to contact a second surface of the ply apart from the corner. A corner displacer is coupled to at least one of the first support and the second support. The corner displacer is configured to displace the corner in a first direction and an opposite second direction to thereby bend portions of the ply and the backing between the corner and the first support and the second support, and the portions of the ply and the backing thereby separate from each other.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a divisional of U.S. application Ser. No. 18/231,326, filed Aug. 8, 2023, which is incorporated herein by reference in its entirety.

FIELD OF THE DISCLOSURE
[0002] The present disclosure generally relates to systems and methods for separating thermoset prepreg fiber-reinforced plastic plies from backings. More specifically, the present disclosure relates to systems and methods for initiating separation of corners of plies from backings.
BACKGROUND
[0003] High-performance aerospace components are typically constructed of thermoset prepreg fiber-reinforced plastic materials, which are also referred to simply as composite materials. Such materials are a mixture of high-strength fibers that are fixedly held in a plastic matrix. The fibers provide most of the strength of the material, and the plastic matrix holds the fibers together and distributes loads between the fibers. The fibers are typically constructed of carbon (more specifically, graphite) or glass. In some cases, composite materials include a combination of fiber materials.
[0004] Prepreg is raw material including a plastic matrix pre-mixed with high-strength fibers, typically provided in long and continuous rolls of thin fabric. Prepreg is typically used by component manufacturers to ensure that the matrix is well-mixed with the fibers and that the matrix-to-fiber ratio is correct.
[0005] Thermoset composite materials typically use a thermoset plastic resin, such as an epoxy, as the matrix. Thermoset plastics are initially sticky, viscous liquids that solidify by undergoing a chemical reaction, referred to as curing, after which they remain solid. In contrast, thermoplastics melt when heated and can be remelted many times.
[0006] Thermoset prepreg material has tack or stickiness until it is cured, so such a material is placed on a backing, such as a plastic-coated paper, during manufacturing. Without a backing, the material does not sufficiently hold its shape, and pieces of the material can stick to each other and themselves. In some cases, a second sheet of material is also placed on the top side of prepreg. The second sheet, referred to as a cover film or polyfilm, is typically a thin plastic sheet.
[0007] Components constructed of prepreg composites are typically manufactured by stacking layers of the raw material, referred to as plies. Such a process is referred to as layup, and the unfinished component may be referred to as a layup during the process. The plies are pre-cut to the shape of the component and stacked, one at a time, on a mold that defines the contours of the component. Components manufactured via such a process may include several plies to several hundred plies, depending on the nature and function of the components. After layup is complete, the components are placed in a pressurized oven that cures the thermoset plastic.
[0008] During layup, the plies are typically cut to the appropriate shape while the composite material is still on its backing, and then the backing is removed from each ply immediately before it is placed on the layup. Initiating separation, or peeling, of the ply and backing is difficult because the resin in the ply sticks aggressively to the backing. Additionally, because the ply and backing are cut to shape together, there is no excess backing protruding past the edge of the ply that could be used to facilitate separation.
[0009] Operators that manually lay up composite components develop personal techniques for separating plies from their backings. Such techniques typically include ruffling, flicking, or striking

a corner of the ply; scraping a corner of the ply with a blade; and/or inserting a sharp blade between the ply and backing at a corner. After a corner of the ply has been separated from its backing, peeling the remainder of the ply from the backing is relatively easy—the separated portions of the ply and the backing are grasped in separate hands and pulled apart.

[0010] Automated layup systems and methods also experience difficulties when separating backings from plies, and the techniques used by human operators are not feasible for automated equipment that lacks the dexterity, visual processing, and adaptability of human operators. Accordingly, improved systems and methods for initiating separation of backings from plies would be beneficial.

SUMMARY

[0011] In an Example 1, a system for separating a ply from a backing is provided. The backing includes a first surface facing away from the ply, the ply includes a second surface facing away from the backing, and the ply and the backing together define a corner. The system includes a first support configured to contact the first surface apart from the corner; a second support configured to contact the second surface apart from the corner; and a corner displacer coupled to at least one of the first support and the second support. The corner displacer is configured to displace the corner in a first direction and an opposite second direction to thereby bend portions of the ply and the backing between the corner and the first support and the second support, and the portions of the ply and the backing thereby separate from each other.

[0012] In an Example 2, the system of Example 1, wherein the corner displacer is configured to bend the portions of the ply and the backing about a bending axis, and the bending axis is substantially parallel to the first surface and the second surface.

[0013] In an Example 3, the system of Example 1, wherein the corner displacer is a first corner displacer, and further including a second corner displacer coupled to the at least one of the first support and the second support.

[0014] In an Example 4, the system of Example 3, wherein the first corner displacer is pivotable relative to the at least one of the first support and the second support about a first axis, the second corner displacer is pivotable relative to the at least one of the first support and the second support about a second axis, and the second axis is different than the first axis.

[0015] In an Example 5, the system of Example 4, wherein the first axis and the second axis are substantially perpendicular.

[0016] In an Example 6, the system of Example 3, wherein the ply and the backing include a first edge and a second edge, the first edge and the second edge intersect at the corner, and the first corner displacer is configured to contact the first edge and the second corner displacer is configured to contact the second edge.

[0017] In an Example 7, the system of Example 1, wherein the second support includes an end effector configured to displace the ply and the backing relative to the first support.

[0018] In an Example 8, the system of Example 1, further including a gripper coupled to the first support, and the gripper is configured to grip the separated portion of the backing.

[0019] In an Example 9, a method for separating a ply from a backing by using a separating system is provided. The backing includes a first surface facing away from the ply, the ply includes a second surface facing away from the backing, and the backing and the ply together define a corner. The separating system includes a first support, a second support, and a corner displacer. The method includes (A) contacting the first support against the first surface and the second support against the second surface such that the corner is disposed apart from the first support and the second support; (B) actuating the corner displacer to displace the corner in a first direction and an opposite second direction to thereby bend portions of the ply and the backing between the corner and the first support and the second support, the portions of the ply and the backing thereby separating from each other; and (C) relatively displacing the separated portions of the ply and the backing away from each other, and other portions of the ply and the backing thereby separating

from each other.
[0020] In an Example 10, the method of Example 9, wherein the corner displacer bends the portions of the ply and the backing about a bending axis, and the bending axis is substantially parallel to the first surface and the second surface.
[0021] In an Example 11, the method of Example 10, wherein the ply includes fibers and a resin, the fibers are elongated in a fiber direction, and the bending axis are substantially perpendicular to the fiber direction.
[0022] In an Example 12, the method of Example 9, wherein an adhesive bond is disposed between the ply and the backing, and bending the portions of the ply and the backing between the corner and the first support and the second support causes the adhesive bond to break, and the portions of the ply and the backing thereby separate from each other.
[0023] In an Example 13, the method of Example 9, wherein actuating the corner displacer to bend the portions of the ply and the backing includes actuating the corner displacer for a plurality of cycles.
[0024] In an Example 14, the method of Example 13, wherein each cycle has a maximum duration of two seconds.
[0025] In an Example 15, the method of Example 9, wherein the corner displacer is a first corner displacer, and further including a second corner displacer, and actuating the corner displacer to displace the corner includes pivoting the first corner displacer and the second corner displacer relative to at least one of the first support and the second support.
[0026] In an Example 16, the method of Example 15, wherein pivoting the first corner displacer and the second corner displacer relative to the at least one of the first support and the second support includes pivoting the first corner displacer about a first axis and pivoting the second corner displacer about a second axis, and the second axis is different than the first axis.
[0027] In an Example 17, the method of Example 16, wherein the first axis and the second axis are substantially perpendicular.
[0028] In an Example 18, the method of Example 15, wherein the ply and the backing include a first edge and a second edge, the first edge and the second edge intersecting at the corner, and pivoting the first corner displacer and the second corner displacer relative to the at least one of the first support and the second support includes contacting the first corner displacer with the first edge and contacting the second corner displacer with the second edge.
[0029] In an Example 19, the method of Example 9, wherein the second support includes an end effector, further including displacing the ply and the backing relative to the first support by using the end effector, and wherein contacting the first support against the first surface and the second support against the second surface includes holding the ply and the backing against the first support by using the end effector.
[0030] In an Example 20, the method of Example 9, wherein the separating system further includes a gripper coupled to the first support, and relatively displacing the separated portions of the ply and the backing away from each other includes gripping the backing with the gripper.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0031] For illustrative purposes, the drawings show embodiments which are presently preferred. However, it should be understood that the present disclosure is not limited to the embodiments illustrated in the drawings.
[0032] FIG. **1** is a perspective view of a system for separating thermoset prepreg fiber-reinforced plastic plies from their backings, according to an embodiment of the present disclosure.
[0033] FIG. **2** is a perspective view of a separator of the system of FIG. **1**.

[0034] FIG. **3** is a detail perspective view of a separation station of the separator of FIG. **2**.
[0035] FIG. **4** is a detail top view of a first support and corner displacers of the separation station of FIG. **3**.
[0036] FIG. **5** is a perspective view of one of the corner displacers of FIG. **4**.
[0037] FIG. **6** is a side view of the corner displacer of FIG. **5**.
[0038] FIG. **7** is a top view of the first support and a gripper of the separation station of FIG. **3**.
[0039] FIG. **8** is a perspective view of the gripper of FIG. **7**.
[0040] FIG. **9** is an opposite perspective view of the gripper of FIG. **8**.
[0041] FIG. **10** is a perspective view of a system for separating plies from their backings, according to another embodiment of the present disclosure.
[0042] FIG. **11** is a flow diagram of a method for separating plies from their backings, according to another embodiment of the present disclosure.
[0043] FIG. **12** is a partial perspective view of a sheet including a ply and a backing to be separated via the method of FIG. **11**.
[0044] FIG. **13** is a detail top view of the first support, the corner displacers, and an end effector of the separation station of FIG. **3** with the sheet of FIG. **12** positioned between the first support and the end effector.
[0045] FIG. **14** is a detail perspective view of the first support, the corner displacers, and the end effector of FIG. **13** with the sheet of FIG. **12** positioned between the first support and the end effector and the corner displacers displacing the corner of the sheet.
[0046] FIG. **15** is a detail top view of the sheet of FIG. **12** specifically illustrating an area of the sheet initially manipulated to by the system of FIG. **1**.
[0047] FIG. **16** is a detail perspective view of the gripper of the system of FIG. **1** gripping the initially manipulated area of the sheet of FIG. **15**.
[0048] FIG. **17** is a detail perspective view of the gripper and the end effector of the system of FIG. **1** separating the ply and the backing of the sheet of FIG. **12**.

DETAILED DESCRIPTION OF THE DRAWINGS

[0049] Referring to the drawings, wherein like reference numerals identify corresponding or similar elements throughout the several views, FIG. **1** illustrates a system **100** for separating thermoset prepreg fiber-reinforced plastic plies from their backings, according to an embodiment of the present disclosure. The system **100** generally includes a frame **102** that carries a separator **104**. The frame **102** generally includes a plurality of elongated beams and columns **106** that may be constructed of various appropriate materials, such as metals, more specifically aluminum, steel, or iron. The frame **102** may be fixedly coupled to a ground surface (for example, via fasteners—not shown). In other embodiments, the frame **102** may be movable relative to the ground surface (for example, by including a plurality of casters—not shown). In some embodiments and as illustrated, the frame **102** movably supports the separator **104** via a linear actuator **108**, more specifically a belt-driven linear actuator, and a linear slide **109**. In other embodiments, the separator **104** may be fixedly supported by the frame **102**. As described in further detail below, the separator **104** includes a first support **110** for contacting and supporting sheets including plies and backings. As also described in further detail below, the separator **104** also includes one or more end effectors **112** that are coupled to and displaced by robotic arms (not shown). The end effectors **112** (1) displace the sheets including plies and backings relative to the remainder of the separator **104**, and (2) act as second supports for contacting and holding the sheets against the first support **110**.
[0050] FIG. **2** illustrates the separator **104** in further detail. The separator **104** generally includes a lower plate **114**, an upper plate **116** that defines the first support **110**, and a plurality of intermediate columns **118** coupling the lower plate **114** and the upper plate **116**. The components may be constructed of various appropriate materials, such as metals, more specifically aluminum, steel, or iron. As described in further detail below, the lower plate **114** and the upper plate **116** couple to a plurality of driven components that interact with sheets including plies and backings. In some

embodiments and as illustrated, the upper plate **116** and the driven components form two separation stations **120** for separately interacting with different sheets including plies and backings. In some embodiments and as illustrated, the separation stations **120** have different orientations, or face different directions, to facilitate placement of plies by the robotic arms and end effectors **112**. In other embodiments, the separator **104** includes a single separation station (for example, the system **100** includes a single, relatively large robotic arm and a single end effector **112**, and/or if the system **100** is used to separate relatively small plies from relatively small backings). In such embodiments, the single separation station may be rotatable relative to the frame **102**. In some embodiments, the separator **104** includes three or more separation stations.

[0051] FIG. **3** illustrates the separator **104** in further detail. FIG. **3** only illustrates one separation station **120** of the separator **104**, although it is understood that the other separation station **120**, respectively, have similar features. The illustrated separation station **120** includes a first corner displacer **122** and a second corner displacer **124**. In other embodiments, the separation station **120** includes a single corner displacer or one or more additional corner displacers. As described in further detail below, the corner displacers **122**, **124** are actuated to repeatedly displace the corner of a sheet secured between the first support **110** and the end effector **112**, acting a second support. This action repeatedly bends portions of the sheet between the corner and the first support **110** and the end effector **112**, and those portions of the ply and backing thereby separate from each other. The separation station **120** further includes a gripper **126**. In other embodiments, the separation station **120** includes one or more additional grippers. As described in further detail below, the gripper **126** receives a sheet after being bent by the corner displacers **122**, **124**. The gripper **126** grips the separated portion of the backing, which facilitates separating the remaining portions of the ply and the backing.

[0052] With continued reference to FIG. **3**, the end effector **112** generally includes an operative end **128** that is configured to couple to a ply of a sheet. In some embodiments and as illustrated, the operative end **128** includes a suction applicator for coupling to the ply of the sheet. In some embodiments, the suction applicator may employ active suction (for example, by coupling to one or more pneumatic components) to couple to the ply of the sheet. In other embodiments, the end effector **112** may be a different type of mechanical device, such as a needle gripper.

[0053] FIG. **4** illustrates the first support **110** and the corner displacers **122**, **124** of the separator **104** in further detail. Upon actuation, the first corner displacer **122** pivots about a first axis **130** relative to the first support **110** and the second corner displacer **124** pivots about a different second axis **132** relative to the first support **110**. In some embodiments and as illustrated, the first axis **130** and the second axis **132** are substantially perpendicular to each other (that is, perpendicular ±5 degrees). In other embodiments, the axes **130**, **132** are not substantially perpendicular to each other. In some embodiments, the first axis **130** and the second axis **132** are substantially parallel (that is, parallel ±5 degrees) to the plane of the first support **110**. In other embodiments, the axes **130**, **132** are not substantially parallel to the plane of the first support **110**.

[0054] FIGS. **5** and **6** illustrate the first corner displacer **122** in further detail. The second corner displacer **124** is not illustrated in further detail, although it is understood that the second corner displacer **124** has similar features as the first corner displacer **122**. The first corner displacer **122** includes a prime mover **134**, more specifically a linear actuator, and even more specifically a pneumatic or hydraulic actuator. In other embodiments the prime mover **134** may take different forms. For example, the prime mover **134** may be a different linear actuator, such as a rack and pinion drive, or a rotary actuator, such as an electric motor. The prime mover **134** is coupled to and displaces, more specifically pivots, an arm **136** of the corner displacer **122** relative to the upper plate **116** (shown elsewhere). Upon being displaced relative to the upper plate **116**, the arm **136** displaces a corner of a sheet secured between the first support **110** and the end effector **112**, acting as the second support (shown elsewhere). The arm **136** includes an operative end **138** that is configured to contact the sheet. In some embodiments and as illustrated, the operative end **138** includes a wing-like shape as viewed from above.

[0055] FIGS. **7-9** illustrate the gripper **126** in further detail. The gripper **126** generally includes a first, or upper, jaw **140** and a second, or lower, jaw **142** that are operable to “sandwich” or grip a backing of a sheet therebetween. The first jaw **140** includes a suction applicator **144** for coupling to the backing of the sheet. In some embodiments, the suction applicator **144** may employ active suction (for example, by coupling to one or more pneumatic components) to couple to the backing of the sheet. In other embodiments, a different type of mechanical device may be used instead of the suction applicator **144**, such as a needle gripper. The first jaw **140** is displaceable, more specifically pivotable, relative to the upper plate **116** (FIG. **7**) via a prime mover **146** (FIGS. **8** and **9**), more specifically a linear actuator, and even more specifically a pneumatic or hydraulic actuator. In other embodiments the prime mover **146** may take different forms. For example, the prime mover **146** may be a different linear actuator, such as a rack and pinion drive, or a rotary actuator, such as an electric motor. The second jaw **142** is similarly displaceable, more specifically pivotable, relative to the upper plate **116** (FIG. **7**) via a prime mover **148** (FIGS. **8** and **9**), more specifically a linear actuator, and even more specifically a pneumatic or hydraulic actuator. In other embodiments the prime mover **148** may take different forms. For example, the prime mover **148** may be a different linear actuator, such as a rack and pinion drive, or a rotary actuator, such as an electric motor.

[0056] FIG. **10** illustrates a system **200** for separating plies from their backings, according to another embodiment of the present disclosure. The system **200** generally includes similar components as the system **100**, although the system **200** only includes a single corner displacer **222**. The corner displacer **222** includes an operative end **228** with a letter T-like shape with a notch **229** for avoiding direct contact with the suction applicator of the end effector **112**.

[0057] FIG. **11** illustrates a flow diagram of a method **300** for separating plies from their backings, according to another embodiment of the present disclosure, and FIGS. **12-17** illustrate certain components and actions associated with the method **300**. The method **300** is illustratively described in connection with the system **100**, although it is understood that the method **300** may similarly be employed in connection with the system **200**. The method **300** begins at block **302** by providing a sheet comprising a ply and a backing. An illustrative sheet **400** is shown in FIG. **12**. The backing **402** of the sheet **400** includes a first surface **404** that faces away from the ply **406**, and the ply **406** includes a second surface **408** that faces away from the backing **402**. Opposite the first surface **404** and the second surface **408**, an adhesive bond **410** is disposed between and couples the ply **406** and the backing **402**. The backing **402** and the ply **406** together define a first edge **412** and a second edge **414** of the sheet **400**, and the first edge **412** and the second edge **414** intersect at a corner **416**. In some embodiments and as illustrated, the corner **416** is a sharp corner. In other embodiments, the corner **416** is a rounded corner. In some embodiments, the first edge **412** and the second edge **414** are substantially perpendicular to each other (that is, perpendicular ±5 degrees). In other embodiments, the first edge **412** and the second edge **414** are not substantially perpendicular to each other. The ply **406** includes a resin **418** that carries a plurality of reinforcing fibers **420**, and the fibers **420** are elongated in a fiber direction. In some embodiments and as illustrated, the fiber direction runs into the corner **416** or, stated another way, substantially bisects (that is, bisect ±5 degrees) the angle between the first edge **412** and the second edge **414**. In other embodiments, the fiber direction is substantially parallel (that is, parallel ±5 degrees) to the first edge **412** or the second edge **414**. In some embodiments, the ply **406** is a cross-ply or a woven ply including two sets of fibers running in substantially perpendicular directions. For such plies, one set of fibers necessarily runs into the corner **416** regardless of the orientation of the fibers relative to the first edge **412** and the second edge **414**.

[0058] With continued reference to FIG. **11** and additional reference to FIG. **13**, the method **300** continues at block **304** by positioning the sheet **400** relative to the separator **104** such that the corner is disposed apart from, or overhangs, the first support **110** and the second support, more specifically, the end effector **112**. In some embodiments, the sheet **400** is positioned by displacing the sheet **400** relative to the first support **110** by using the end effector **112**. In some embodiments, the first support **110** contacts the first surface **404** of the backing **402** (shown elsewhere) and the second support, more specifically, the end effector **112**, contacts the second surface **408** of the backing **402** (shown elsewhere). As a result, the sheet **400** is “sandwiched” between the first support **110** and the second support, more specifically, the end effector **112**.

[0059] With continued reference to FIG. **11** and additional reference to FIGS. **13** and **14**, the method **300** continues at block **306** by actuating the corner displacers **122**, **124** to repeatedly displace the corner **416** of the sheet **400** in a first direction (for example, upward) and an opposite second direction (for example, downward), repeatedly bend areas of the sheet **400** adjacent to the corner **416**, and thereby initiate separation of the ply **406** and the backing **402** (shown elsewhere). In some embodiments, bending the areas of the sheet **400** adjacent to the corner **416** causes the adhesive bond **410** (shown elsewhere) to break, thereby initiating separation of the ply **406** and the backing **402**. In some embodiments, the first corner displacer **122** contacts the first edge **412** of the sheet **400** and the second corner displacer **124** contacts the second edge **414** of the sheet **400**. In some embodiments, the corner displacers **122**, **124** cause the sheet **400** to bend about a bending axis **422** that is substantially parallel (that is, parallel ±5 degrees) to the first surface **404** and the second surface **408** of the sheet **400** (shown elsewhere). In some embodiments and as illustrated, the bending axis **422** is substantially perpendicular (that is, perpendicular ±5 degrees) to the fiber direction of the ply **406**. In some embodiments, more specifically those in which the fiber direction of the ply **406** is substantially parallel to the first edge **412** or the second edge **414** of the sheet **400**, the first corner displacer **122** or the second corner displacer **124**, respectively, may not be actuated, and the first corner displacer **122** and the second corner displacer **124** may not contact the first edge **412** or the second edge **414**, respectively. In some embodiments, the bending axis **422** moves during different portions of each bending cycle. More specifically, during the portion of the cycle in which the sheet **400** bends upward, the bending axis **422** may be aligned with the portion of the end effector **112** nearest to the corner **416**, and during the portion of the cycle in which the sheet **400** bends downward, the bending axis **422** may be aligned with the portion of the first support **110** nearest to the corner **416**. In some embodiments, the corner displacers **122**, **124** are actuated, and the sheet **400** is thereby bent, for at least one cycle, and more preferably two or three cycles. In some embodiments, each cycle has a maximum duration of two seconds.

[0060] Referring specifically to FIG. **15**, the first corner displacer **122** (shown elsewhere) contacts a first line segment **424** of the first edge **412** of the sheet **400** and the second corner displacer **124** (shown elsewhere) contacts a second line segment **426** of the second edge **414** of the sheet **400**. The first line segment **424**, the second line segment **426**, and an imaginary line segment **428** connecting the ends of the first line segment **424** and the second line segment **426** opposite the corner **416** bound an initially manipulated area **430** of the sheet **400**. In some embodiments and as illustrated, the end effector **112** may be at least partially disposed in the initially manipulated area **430**. In some embodiments, the system **100** may specifically interact with the initially manipulated area **430** in the manner described below.

[0061] After bending the sheet **400**, the sheet **400** is displaced toward the gripper **126**, for example, via the end effector **112**. With reference again to FIG. **11**, the method continues at block **308** by relatively displacing the ply **406** and the backing **402** in the initially manipulated area **430** of the sheet **400** away from each other. This action causes other portions of the ply **406** and the backing **402** to separate from each other, thereby completing separation of the ply **406** and the backing **402**. More specifically and as shown in FIG. **16**, the first jaw **140** of the gripper **126** may first couple to the backing **402**, via the suction applicator **144**, and the first jaw **140** may then pivot away from the end effector **112**. Next and as shown in FIG. **17**, the second jaw **142** may then pivot toward the backing **402** and the first jaw **140** to secure the backing between the jaws **140**, **142**. In some embodiments, the first jaw **140** and the second jaw **142** secure the backing **402** in the initially manipulated area **430** of the sheet **400**. The end effector **112** then moves away from the gripper **126** to separate the ply **406** and the backing **402** to complete separation of the ply **406** from the backing **402**. In some embodiments, the end effector **112** then displaces the ply **406** to a layup or another appropriate location.

[0062] While exemplary embodiments have been illustrated, it is understood that the present disclosure is not limited thereto. The embodiments may be changed, modified and further applied by those skilled in the art. Therefore, the present disclosure is not limited to the details shown and described, but also includes all such changes and modifications.

Claims

1. A method for separating a ply from a backing by using a separating system, the backing comprising a first surface facing away from the ply, the ply comprising a second surface facing away from the backing, and the backing and the ply together defining a corner, the separating system comprising a first support, a second support, and a corner displacer, the method comprising: contacting the first support against the first surface and the second support against the second surface such that the corner is disposed apart from the first support and the second support; actuating the corner displacer to displace the corner in a first direction and an opposite second direction to thereby bend portions of the ply and the backing between the corner and the first support and the second support, the portions of the ply and the backing thereby separating from each other; and relatively displacing the separated portions of the ply and the backing away from each other, and other portions of the ply and the backing thereby separating from each other.

2. The method of claim 1, wherein the corner displacer bends the portions of the ply and the backing about a bending axis, the bending axis being substantially parallel to the first surface and the second surface.

3. The method of claim 2, wherein the ply comprises fibers and a resin, the fibers being elongated in a fiber direction, and the bending axis being substantially perpendicular to the fiber direction.

4. The method of claim 1, wherein an adhesive bond is disposed between the ply and the backing, and bending the portions of the ply and the backing between the corner and the first support and the second support causes the adhesive bond to break, the portions of the ply and the backing thereby separating from each other.

5. The method of claim 1, wherein actuating the corner displacer to bend the portions of the ply and the backing comprises actuating the corner displacer for a plurality of cycles.

6. The method of claim 5, wherein each cycle has a maximum duration of two seconds.

7. The method of claim 1, wherein the corner displacer is a first corner displacer, and further comprising a second corner displacer, and actuating the corner displacer to displace the corner comprises pivoting the first corner displacer and the second corner displacer relative to at least one of the first support and the second support.

8. The method of claim 7, wherein pivoting the first corner displacer and the second corner displacer relative to the at least one of the first support and the second support comprises pivoting the first corner displacer about a first axis and pivoting the second corner displacer about a second axis, the second axis being different than the first axis.

9. The method of claim 8, wherein the first axis and the second axis are substantially perpendicular.

10. The method of claim 7, wherein the ply and the backing include a first edge and a second edge, the first edge and the second edge intersecting at the corner, and pivoting the first corner displacer and the second corner displacer relative to the at least one of the first support and the second support comprises contacting the first corner displacer with the first edge and contacting the second corner displacer with the second edge.

11. The method of claim 1, wherein the second support comprises an end effector, further comprising displacing the ply and the backing relative to the first support by using the end effector, and wherein contacting the first support against the first surface and the second support against the second surface comprises holding the ply and the backing against the first support by using the end effector.

12. The method of claim 1, wherein the separating system further comprises a gripper coupled to the first support, and relatively displacing the separated portions of the ply and the backing away from each other comprises gripping the backing with the gripper.